A Decision Support System for Prioritising Industrial Maintenance Interventions

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Abstract. Industrial plants often have more pending maintenance interventions than technicians, spare parts and downtime windows can absorb, so their order has to be decided. Following the Kitchenham guidelines and the PRISMA 2020 statement, 554 records from IEEE Xplore and Web of Science (2021–2026) were screened and 31 studies with evaluation in a real setting were analysed for their criteria, methods, system architecture and validation. Resource constraints, failure likelihood and production impact dominate the criteria, while safety is explicit in only seven studies and is usually traded off against cost. Few proposals are implemented systems, priorities are rarely updated or explained, and only six were tested in the plant. Based on these gaps, we propose a decision support system that combines machine learning failure prediction with AHP and TOPSIS, treats safety as a non-compensable constraint and explains each position in the ranking.

Keywords: Industrial maintenance · Decision support systems · Maintenance prioritisation · Multi-criteria decision-making · Predictive maintenance.

1 Introduction

Maintenance managers rarely face a single failure at a time. On a typical day, corrective work orders, preventive tasks and alerts from condition monitoring compete for the same technicians, spare parts and production stoppages, and the order in which they are handled affects safety, output and cost. In many plants this order is still set by experience or by simple urgency codes in the maintenance management system, which makes it hard to justify and to apply consistently across shifts and departments [1, 14].

The spread of sensors and machine learning in Industry 4.0 has improved the ability to predict failures, but a prediction does not tell the manager which intervention to perform first when several are pending. Turning failure predictions, costs, resources and expert knowledge into a justified order of interventions is a multi-criteria decision problem, and decision support systems (DSS) are a natural way to address it.

This paper has two objectives. The first is to review how recent studies prioritise industrial maintenance interventions: which criteria they use, which methods turn those criteria into a priority order, how the resulting systems are built and how they were validated. The second is to use the gaps found in that review to propose the conceptual architecture of a DSS for this problem.

The paper is organised as follows. Section 2 describes the review method and Section 3 introduces the main maintenance strategies. Section 4 analyses the prioritisation criteria and methods, Section 5 the architecture and validation of the systems, and Section 6 compares the approaches and identifies gaps and trends. Section 7 presents the proposed system and Section 8 concludes.

2 Review Methodology

This review follows a structured state-of-the-art approach based on the guidelines of Kitchenham and Charters [17] and reports the study selection according to PRISMA 2020 [21]. It is not a full systematic literature review, since no formal quality scoring was performed; the robustness of each study was instead assessed through the validation and limitations recorded during data extraction. Four research questions (RQ) guided the review:

- **RQ1:** Which criteria are considered when prioritising industrial maintenance interventions?
- **RQ2:** Which methods (multi-criteria decision-making, optimisation, machine learning) are used to prioritise these interventions?
- **RQ3:** How are maintenance decision support systems designed in terms of architecture, data sources and interaction with the decision-maker?
- **RQ4:** How are the proposed approaches validated, and what limitations do they present?

2.1 Search Strategy and Eligibility Criteria

IEEE Xplore and the Web of Science Core Collection were searched; Scopus was not available through the institutional subscription. The search string combines three groups of concepts: maintenance, prioritisation and decision-making, and the industrial context. A first version, which applied all terms to every field, returned about 1500 records in IEEE Xplore, many of them only because journal titles matched the industrial terms. The final version therefore requires *maintenance* in the title and the other concepts in the abstract:

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IEEE Xplore: ("Document Title":maintenance) AND ("Abstract":prioriti* OR
"Abstract":priority OR "Abstract":criticality OR "Abstract":"decision support" OR
"Abstract":"multi-criteria" OR "Abstract":AHP OR "Abstract":TOPSIS) AND
("Abstract":industr* OR "Abstract":manufactur* OR "Abstract":equipment)

Web of Science: TI=(maintenance) AND AB=(prioriti* OR priority OR criticality OR
"decision support" OR "multi-criteria" OR AHP OR TOPSIS) AND AB=(industr* OR
manufactur* OR equipment) AND PY=(2021-2026)
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Both searches were run on 19 October 2026 and limited to 2021–2026; one study published online in 2026 appears in a 2027 journal issue [18]. In Web of Science, results were limited to articles, proceedings papers and early access items in English, within ten engineering, operations research, management and computer science categories, which reduced them from 582 to 385. IEEE Xplore returned 169 records.

A study was included if it was a peer-reviewed journal or conference paper, published between 2021 and 2026, written in English, proposing or applying a prioritisation or decision method for maintenance interventions in an industrial context, and presenting some form of evaluation. Studies were excluded for any of the reasons in Table 1. Criterion E8 was added during screening, and E9 and E10 were defined after data extraction and before synthesis, to focus the review on its central problem and on evidence from real settings. Both were applied in the same way to every study read in full.

Table 1. Exclusion criteria.

Code	Criterion
E1	Duplicate record.
E2	Not a peer-reviewed primary study (review, survey, editorial or preprint).
E3	Outside the industrial context (e.g., healthcare, buildings, civil infrastructure, IT maintenance).
E4	Only failure prediction, detection or diagnosis, with no decision component.
E5	No evaluation of any kind.
E6	Insufficient methodological detail.
E7	Full text not accessible.
E8	Maintenance management without decisions on interventions or assets (e.g., skills, barriers, KPIs).
E9	The decision is not the prioritisation, selection or scheduling of interventions or assets (e.g., strategy selection, interval optimisation, joint production and maintenance scheduling).
E10	Not applied to real data or an industrial case with a comparison against current practice, another method or a before/after measurement.

2.2 Study Selection and Data Extraction

Fig. 1 summarises the selection. Of the 554 records retrieved, 45 duplicates were removed by DOI and normalised title. The remaining 509 were screened by title and abstract, a stage that excluded 233 records, mostly under E8 ($n = 108$) and E3 ($n = 59$). Of the 276 reports sought, 47 could not be accessed (E7). After full-text reading of the remaining 229, a further 198 were excluded, mainly under E9 ($n = 60$) and E10 ($n = 62$), leaving 31 studies.

For each included study, an extraction form recorded the sector, the prioritisation criteria (RQ1), the decision method and the origin of the weights (RQ2), the architecture and data sources (RQ3), the validation design and results, the limitations (RQ4) and whether the system explains its recommendations. A large language model produced a first classification at screening and a first version of each form; every decision and value was then checked against the original article by one of the authors.

2.3 Limitations of the Review

Only two databases were searched, and requiring *maintenance* in the title may have missed studies that mention it only in the abstract. Defining E9 and E10 after extraction introduces a risk of selection bias, although both were applied uniformly and are documented in Table 1. Finally, 47 reports could not be retrieved, and the use of a language model was mitigated by having every output verified by the authors.

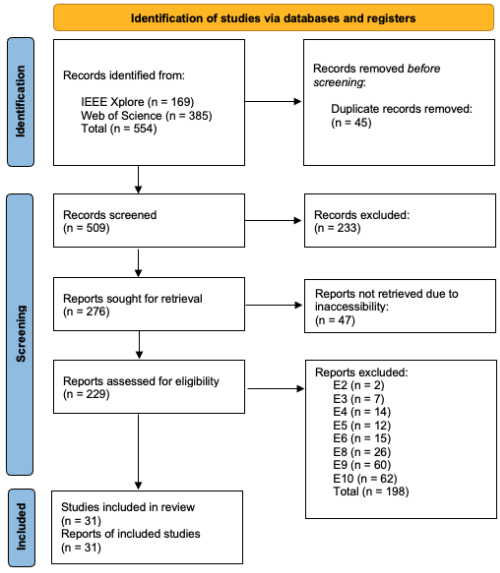


Fig. 1. PRISMA 2020 flow diagram of the study selection.

3 Background: Maintenance Strategies

Maintenance strategies differ in what triggers an intervention and in the information they require (Table 2). **Corrective** maintenance restores equipment after a failure; it needs no planning but exposes the plant to unplanned downtime, secondary damage and safety risks, so it suits only equipment whose failure has small consequences. **Preventive** maintenance acts at fixed time or usage intervals, which makes work easy to plan, but the intervals reflect average behaviour rather than the state of each machine, so some components are replaced too early and others fail before their turn. **Condition-based** maintenance (CBM) intervenes when a measured indicator, such as vibration or temperature, crosses a threshold. **Predictive** maintenance goes further and estimates the remaining useful life (RUL) with data-driven, physics-based or hybrid models. A RUL estimate, however, does not say which of several pending interventions should come first, nor how to weigh failure likelihood against safety, production or cost; many retrieved studies stopped at this stage and were excluded under E4. **Prescriptive** maintenance closes this gap by recommending the action, its timing and its resources, combining predictions with costs, constraints and expert knowledge. The DSS proposed in this work moves in this direction, turning failure predictions into a justified order of interventions.

Two management frameworks are often combined with these strategies. Reliability-Centred Maintenance (RCM) defines the policy of each asset through an analysis of its functions, failure modes and consequences; its tools, Failure Mode and Effects Analysis (FMEA) and criticality analysis, introduced the ranking of assets by severity and likelihood that underlies many of the models in Section 4. Total Productive Mainte-

nance (TPM) involves operators in equipment care and measures performance through Overall Equipment Effectiveness (OEE), bringing the production cost of downtime into maintenance decisions.

Table 2. Comparison of maintenance strategies.

Strategy	Trigger	Information required	Main limitation
Corrective	Occurrence of a failure	Failure reports	Unplanned downtime and secondary damage
Preventive	Fixed time or usage interval	Manufacturer data, failure history	Interventions on healthy equipment
Condition-based	Indicator exceeds a threshold	Condition monitoring data	No anticipation of the failure time
Predictive	Estimated RUL	Sensor data, prognostic models	Does not define which action to take or in which order
Prescriptive	Recommended action and timing	Predictions, costs, resources and constraints	Requires integration of data, models and organisational knowledge

4 Prioritisation Criteria and Methods

This section addresses RQ1 and RQ2: the criteria used to rank maintenance interventions in the 31 included studies and the methods that turn them into a priority order or a schedule.

4.1 Prioritisation Criteria

The criteria were grouped into eight categories (Table 3). Only criteria that actually enter the decision model were counted; several authors mention factors they could not use because the data did not exist [13, 25, 28–30].

Table 3. Prioritisation criteria used in the included studies (n = number of studies).

Category	Typical indicators	Studies	*n*
Resources and operational constraints	Crews, skills, man-hours, budget, spare parts, downtime windows, travel	[4–7,10–13,18–20,22,24–26,29–31,33]	19
Likelihood of failure	Occurrence, failure rate, MTBF, number of past failures	[1,2,5–9,11,14,15,19,20,22,23,25,33]	16
Production or service impact	Downtime, bottlenecks, OEE, line efficiency, energy not supplied, SAIFI, SAIDI	[3,6,9–11,13–16,19,22,25,27,30,32,33]	16
Cost	Intervention cost, downtime and interruption cost, financial risk	[2,3,6,7,9–11,13,19,26,27,29–31,33]	15
Condition, degradation and RUL	Sensor-based health indices, wear, corrosion, predicted RUL	[4,6,7,10,13,16,24–30,33]	14
Detectability	Detection of the failure before it causes damage	[5,8,11,15,19,20,22,23,25]	9
Asset importance	Criticality class, importance measures, regulatory requirements	[1,9,10,12,13,16,19,24,30]	9
Safety and environment	Risk to people, explosion, environmental damage	[2,8,11,15,22,25,27]	7

The most frequent category is not a property of the equipment but the resources available to maintain it: in 19 studies, priority depends on crews, budget, man-hours or downtime windows, so prioritisation is often treated as the allocation of scarce resources rather than as a ranking in isolation. Failure likelihood and production or service impact follow, with 16 studies each; in power networks, impact is measured through SAIFI, SAIDI or energy not supplied and then converted into a cost [3,11,19,27,33]. Condition-based criteria appear in 14 studies, all published from 2022 onwards. Safety is the least represented category: it is explicit in only seven studies, while in others it is a level within a broader "consequences" criterion [5, 20] or is deliberately excluded [32].

4.2 Criticality Analysis and FMEA

Criticality analysis ranks assets or failure modes by combining severity and likelihood, usually in a matrix with a few classes. FMEA adds detectability and summarises each failure mode with the Risk Priority Number, $RPN = S \times O \times D$. The RPN is familiar to maintenance teams, but different score combinations give the same value, ordinal scales are multiplied as if they were ratio scales, the three factors weigh the same, and cost and resources are ignored. The classic form is still used in practice [8,15]. Most other studies modify it: they weight S, O and D through the Analytic Hierarchy Process (AHP) or the Best-Worst Method (BWM) [11,23,25], replace the product by a multi-criteria decision-making (MCDM) ranking [11,19,22,33], represent the uncertainty of expert scores with fuzzy sets [2,23,25], or update the risk over time, after each maintenance action [23] or as a preventive task is delayed [27]. Hedayatnia et al. [16] cluster a condition index and an importance index into nine priority segments, a data-driven version of the criticality matrix.

4.3 Multi-Criteria Decision-Making Methods

MCDM methods serve two tasks. For **weighting**, AHP derives weights from pairwise comparisons and checks their consistency [5, 20, 22, 33]; the check matters, since only four of sixteen expert questionnaires were consistent in the group AHP of Rios et al. [25]. BWM, which needs fewer comparisons, was used mainly in power distribution [3,11,19]. Other studies assume equal weights [7,8,29], tune them against validation metrics [28] or do not report them [11, 30]. For **ranking**, TOPSIS, which orders alternatives by their closeness to an ideal solution, is the most frequent method [3, 5, 19, 20], followed by PROMETHEE II [20, 22]; grey relational analysis [11] and MABAC [28] appear once, and weighted sums are used for simple indices [7,16,29]. VIKOR and ELECTRE were not used. Fuzzy variants, from fuzzy AHP and BWM to cloud models and picture or spherical fuzzy sets [2, 3, 22, 23, 30], capture vague judgements and disagreement between experts, at the cost of many parameters fixed by the authors.

4.4 Optimisation, Scheduling and Machine Learning

A ranking does not say which interventions fit the available resources or when to perform them. Fifteen studies add an optimisation model. Mixed-integer linear programming (MILP) assigns strategies to failure modes under budget, outage and crew

constraints [11, 19], schedules asset fleets [26], allocates technicians [12] or plans over a rolling horizon [6]; smaller integer models select what to repair within a man-hour or budget limit [7, 24, 25]. Larger problems use metaheuristics: adaptive large neighbourhood search [29, 31], particle swarm optimisation [4, 9], a multi-objective genetic algorithm [10] and a two-phase time-window algorithm [13]. Li et al. [18] are the only authors to use deep reinforcement learning (DRL). Priority enters these models either as a coefficient of the objective, which allows trade-offs [6, 7, 25], or as a constraint that guarantees an order at the cost of efficiency [9, 10, 12, 13].

Machine learning (ML) mostly estimates an input to the prioritisation, such as failure probability, RUL or a risk score [6, 7, 10, 24, 28, 30]. Astepe and Alkara [1] instead learn the priority directly from past expert decisions, which avoids explicit weights but reproduces any bias in those decisions. Finally, two studies rank machines by their effect on production, through sensitivity analysis on a line simulation [32] or bottleneck detection on manufacturing execution system (MES) data [14]. Across the sample, methods are rarely used alone: a typical study chains weighting, ranking and, increasingly, prediction or optimisation.

5 Decision Support Systems for Maintenance Prioritisation

This section addresses RQ3, on architecture, data and interaction with the decision-maker, and RQ4, on validation.

5.1 Architecture, Data and Interaction

A DSS combines data, models and an interface through which the decision-maker uses the results. The included studies cover these components to very different degrees. Only nine deliver an implemented tool or pipeline, such as an offshore painting prototype with 3D visualisation [24], a MATLAB tool linking diagnosis, prediction and prescription [30], or an end-to-end pipeline from telemetry ingestion to a MILP schedule [6]; the others at this level are [1, 7, 12, 18, 28, 29]. Thirteen propose a framework or layered architecture without a complete system [2, 3, 8–10, 13–15, 19, 22, 25, 27, 33], and nine contribute only a method [4, 5, 11, 16, 20, 23, 26, 31, 32]. Described architectures follow the same layers (data acquisition, analysis, decision and presentation), and only five studies include a feedback loop or rolling horizon that revises priorities after interventions [6, 8, 14, 15, 25].

The data come from expert judgement, computerised maintenance management system (CMMS) and enterprise resource planning (ERP) records, MES and PLC production data, condition monitoring and engineering models such as grid simulators. Only three studies use public benchmarks [6, 28, 31], case data are often confidential [6, 18, 25, 30], and some models rely on assumed parameters [10, 13, 26]. Integration with the CMMS is recommended but rarely achieved [8, 11], and data quality is a recurring obstacle: manually recorded CMMS data underestimated stoppages compared with MES data [14], and historical priorities used for training contained errors [1].

Interaction with the decision-maker receives little attention. The output is usually a ranked list, a criticality class or a schedule, at most shown as a Gantt chart or a

two-dimensional map [12, 13, 16]. Only Rios et al. [24] let the user compare planning strategies, and acceptance was assessed through focus groups in two studies [14, 24]. Explainability, assessed during extraction, is full in eleven studies [2, 3, 6, 7, 14, 15, 22, 25–27, 33], partial in seventeen and absent in three [4, 18, 31]. It usually comes from the transparency of the method (published weights, rules or objective terms) rather than from a mechanism designed for the user; only Danach et al. [6] link each scheduled intervention to the sensor variables behind it, through SHAP values. Experts take part only in eliciting weights [19, 22, 23, 25], and no system supports a group decision while the plan is made.

5.2 Validation and Industrial Application

All 31 studies use real industrial data, as required by E10, mainly from energy (11 studies), manufacturing (11) and oil and gas (6), with single cases in construction equipment, mining and aviation. Table 4 groups them by type of validation.

Table 4. Types of validation in the included studies (n = number of studies).

Type	Description	Studies	n
Field implementation	Recommendations applied in the plant; indicators compared before and after	[5, 8, 15, 16, 20, 30]	6
Comparison with current practice	Proposed plan compared with the company's plan or with expert decisions on the same data	[1, 2, 4, 11, 12, 18, 19, 24, 25, 27, 29]	11
Comparison with other methods	Same case solved with alternative methods or policies	[3, 6, 7, 13, 22, 23, 28, 33]	8
Simulation or benchmark	Simulation model of the real system or public benchmark instances	[9, 10, 14, 26, 31, 32]	6

Only six studies report results after applying the recommendations in the plant, and the evidence is limited: a year without failures in critical hydroelectric equipment [5, 20], indicators over a six-month pilot [8], higher availability in a cement mill [15], one executed scenario [30] and three inspected circuit breakers [16]. None uses a control group, so the improvement cannot be separated from other changes. The largest group compares the proposal with current practice on the same data; Rios et al. [25], for example, allocate about twice as much effort to critical assets with the same man-hours, and Wehbi et al. [29] reduce total service time by about 42%. These comparisons are retrospective, so the effect on failures and costs remains an estimate. Comparisons between methods often use weak baselines, such as first-fault-first-repair [13] or a purely reactive policy [6], and statistical tests are rare [14, 18, 28]. Although all studies use real data, few provide strong evidence that prioritisation improved maintenance outcomes.

6 Critical Analysis, Gaps and Trends

6.1 Comparison of the Approaches

The studies fall into five families according to the method at the core of the prioritisation (Table 5). None is better in every respect. MCDM and risk-based methods are transparent

and work with expert knowledge alone, but they produce a static ranking that ignores resources unless coupled with optimisation. Optimisation handles resources and time but usually receives the priorities as given. Hybrid approaches use condition data but depend on large histories and add components the user cannot inspect.

Table 5. Comparison of the families of approaches found in the included studies.

Family	Strengths	Limitations	Studies
MCDM	Transparent; uses expert knowledge; needs no sensor data	Static ranking; subjective weights; compensatory aggregation; resources only when combined with optimisation	[3, 5, 11, 19, 20, 22, 25]
Risk and FMEA	Familiar to maintenance teams; consistent with RCM; links severity and likelihood	Ordinal scores; ranks risks rather than interventions; resources rarely considered	[2, 16, 23, 27, 33]
Hybrid with ML	Uses condition data; can be updated as new data arrive	Needs large labelled histories; labels are proxies or past decisions; predictive part hard to explain	[1, 6, 7, 24, 28–30]
Optimisation and scheduling	Handles crews, budgets and time windows; scales to large instances	Priorities given a priori or hidden in costs; many assumed parameters; metaheuristics and DRL hard to explain	[4,9,10,12,13,18,26,31]
Simulation and industrial practice	Grounded in production impact; implemented in real plants	Specific to one line or plant; hard to generalise; evaluation without control	[8, 14, 15, 32]

6.2 Gaps

Six gaps emerge from the analysis, each pointing to a requirement for a decision support system.

G1. Safety is rarely a separate condition. It is explicit in only seven studies and is usually aggregated with other factors [8, 22], so a low safety score can be offset elsewhere. A DSS should treat safety and regulatory requirements as non-compensable constraints.

G2. Ranking and scheduling are separate. Most MCDM and risk-based studies ignore crews, parts and downtime windows, while some scheduling models take priorities as given [12]; few chain both [6, 7, 19, 25]. A DSS should derive priorities and plan interventions in the same process.

G3. Priorities are rarely updated. Only five studies revise priorities through a feedback loop or rolling horizon. A DSS should recompute them when new condition data, failures or work orders appear.

G4. Weights lack a solid basis. They are assumed equal, unreported or taken from few experts [8, 20, 30], and sensitivity analysis is rare [3, 7, 25, 29]. A DSS should make weights explicit, editable and tested for sensitivity.

G5. Explanation and collaboration are limited. Twenty studies explain their recommendations only partially or not at all, only one links each decision to its data [6], and no system supports a joint decision between maintenance, production and safety. A DSS should justify each priority by criterion and let several stakeholders review the plan.

G6. Evidence of real impact is weak. Only six studies applied their results in the plant, always without a control group. A DSS should record decisions and outcomes so that its effect can be measured over time.

6.3 Trends

Nine of the 31 studies date from 2026, including five of the seven hybrid ML studies, against one in 2021 [1]. Four trends stand out: prediction feeding optimisation directly [6, 7, 10, 29]; prescriptive systems that generate and rank alternative actions [30]; explainability through SHAP values or decomposed risk scores [6,28]; and digital twins and IoT architectures as the data layer [9, 10], together with learning-based scheduling [18]. The most recent proposals are also the most complex, and none has yet been evaluated over a long period of real operation.

7 Proposal

We propose a decision support system that helps the maintenance manager decide the order in which pending interventions should be carried out. The system combines the predicted failure probability of each asset with a multi-criteria decision method and returns a justified ranking. Its design follows the gaps identified in Section 6: safety is a non-compensable constraint (G1), criteria weights are elicited from several stakeholders through a structured method and can be tested for sensitivity (G4), and every position in the ranking is explained in terms of the criteria (G5). The ranking is recomputed when new condition data or failures arrive, which partly addresses G3, and the history of decisions is stored so that their outcomes can later be assessed (G6). Joint ranking and scheduling of interventions (G2) is left for future work.

7.1 System Context

- **Target users.** The main user is the maintenance manager or planner. Production and safety managers take part in defining the importance of the criteria.
- **Type of problem.** Semi-structured: part of the information is quantitative (condition data, costs, repair times) and part depends on expert judgement.
- **Synchronous and asynchronous decision.** The ranking is produced asynchronously, at the start of each shift, and recomputed whenever an urgent failure is reported. The definition of weights is a synchronous group activity, repeated when priorities in the plant change.
- **Criteria.** Safety risk is mandatory and cannot be relaxed: interventions with a risk to people are always placed first. Failure probability, production impact, cost, intervention duration and availability of technicians and spare parts can be relaxed and compensate each other.
- **Social aspects.** Maintenance, production and safety staff have different priorities. Each states its own judgements, and the system aggregates them and shows where they diverge.

- **Sensors.** Condition data such as temperature, vibration and torque feed the failure prediction. The prototype will use a public predictive maintenance dataset; validation with real plant data is planned as future work.
- **Interface.** A web application showing the ranking, the explanation of each position and a what-if view where the user changes the weights and sees the effect on the ranking.
- **Technologies.** Machine learning to estimate failure probability, AHP to derive the weights, and TOPSIS to rank the interventions.

7.2 Requirements

Table 6 lists the functional (FR) and non-functional (NFR) requirements and, where applicable, the gap from Section 6.2 that each one addresses.

Table 6. Functional and non-functional requirements of the proposed system.

ID	Requirement	Gap
FR01	Import pending interventions, equipment data and work orders (e.g., from CSV files or a CMMS).	–
FR02	Estimate the failure probability of each asset from its condition data.	–
FR03	Let each stakeholder define the importance of the criteria through pairwise comparisons, warn when judgements are inconsistent ($CR \geq 0.1$) and aggregate the judgements of all stakeholders.	G4, G5
FR04	Place every intervention with a safety risk at the top of the list, regardless of the other criteria.	G1
FR05	Rank the remaining pending interventions and recompute the ranking when new data or failures arrive.	G3
FR06	Explain each position by showing the contribution of each criterion.	G5
FR07	Show how the ranking changes when the weights are changed (sensitivity analysis).	G4
FR08	Store each ranking, the decision taken and, later, its outcome.	G6
NFR01	Performance: produce the ranking in a few seconds for a few hundred interventions.	–
NFR02	Usability: be understandable by users without knowledge of MCDM methods.	–
NFR03	Transparency: no recommendation is shown without its justification.	G5
NFR04	Interoperability: import data in common formats and allow connection to a CMMS.	–
NFR05	Security: require authentication, with separate profiles for managers and other users.	–

7.3 Conceptual Architecture

The architecture (Fig. 2) has five modules. The **data module** collects condition data, work orders, costs and stock levels. The **predictive module** estimates the failure probability of each asset with a machine learning model. The **weighting module** collects the pairwise comparisons of each stakeholder, checks their consistency and aggregates them into a single set of AHP weights. The **decision module** first separates the interventions that involve a safety risk and then ranks all interventions with TOPSIS, keeping the safety-critical ones at the top. The **interface** presents the ranking, the explanation of each position and the sensitivity analysis, and records the decision taken, which feeds back into the data module.

These choices respond directly to the review. Keeping the predictive and decision modules separate allows the prediction model to be replaced without changing the

prioritisation logic, as in the hybrid studies of Section 4. AHP was chosen because it is widely used in the reviewed studies, checks the consistency of judgements and supports group elicitation; the consistency warning in FR03 follows the low share of consistent expert judgements reported in Section 4.3. TOPSIS was chosen because it is the most frequent ranking method in the sample and its result is easy to explain as the closeness of each intervention to an ideal one. Treating safety as a constraint rather than a weight prevents a risk to people from being offset by gains in other criteria, which is the main weakness found in the compensatory models reviewed.

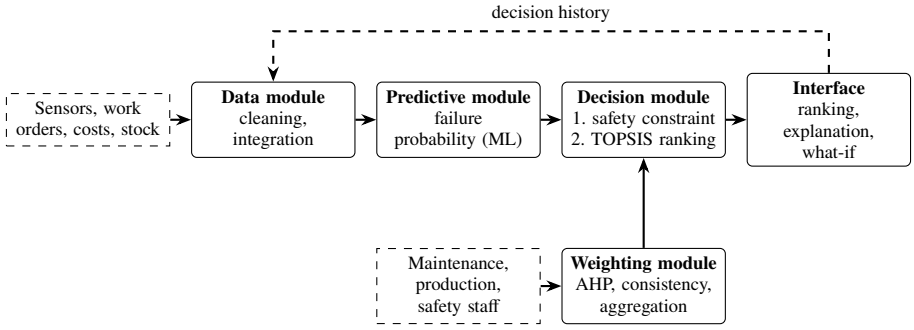


Fig. 2. Conceptual architecture of the proposed system.

8 Conclusions

This review analysed 31 studies, published from 2021 onwards, that prioritise industrial maintenance interventions and evaluate their approach in a real setting. The literature already offers mature building blocks. Criteria are well identified, with resources, failure likelihood and production impact at the centre. Weighting and ranking methods such as AHP, BWM and TOPSIS are widely used, optimisation models handle crews, budgets and time windows, and machine learning provides increasingly reliable failure predictions. What remains open is how these blocks are put together. Safety is seldom treated as a separate condition, ranking and scheduling are usually solved apart, priorities are computed once and rarely updated, weights have a weak basis, recommendations are only partly explained, and few studies show that following the system actually improved maintenance results.

The proposed DSS addresses part of these gaps by combining failure prediction with AHP and TOPSIS, placing safety outside the trade-off between criteria and explaining every position in the ranking. The next steps are to implement the prototype on a public predictive maintenance dataset, elicit the weights with maintenance, production and safety staff, and evaluate the system in scenarios: comparing its ranking with that of an experienced planner, checking how the ranking reacts to changes in the weights, and measuring whether users understand and accept the explanations. Integration with scheduling and validation with real plant data are left for later stages.

In our view, the role of DSS in industrial maintenance will be less about replacing the planner and more about making decisions explicit. As digital twins and condition monitoring make more data available, the value of these systems will lie in showing why an intervention comes first, in letting different departments agree on that order, and in keeping a record that allows the plant to learn from its own decisions.

References

1. Astepe, Ö.Y., Alkara, A.S.: Predicting "Maintenance Priority" with AI. In: 2021 International Conference on Electrical, Communication, and Computer Engineering (ICECCE). IEEE (2021). https://doi.org/10.1109/ICECCE52056.2021.9514218
2. Bahadir, C., Kahraman, C.: Criticality Analysis of Refinery Assets Using Picture Fuzzy Inference System in Reliability Centered Maintenance. International Journal of Fuzzy Systems **28**(5), 1767–1781 (2026). https://doi.org/10.1007/s40815-025-02069-y
3. Bahrami, S., Rastegar, M., Dehghanian, P.: An FBWM-TOPSIS Approach to Identify Critical Feeders for Reliability Centered Maintenance in Power Distribution Systems. IEEE Systems Journal **15**(3), 3893–3901 (2021). https://doi.org/10.1109/JSYST.2020.3014649
4. Bi, S., Li, Z.: Research on maintenance scheme model of mine pipeline and equipment based on particle swarm optimization algorithm. In: 2022 37th Youth Academic Annual Conference of Chinese Association of Automation (YAC). IEEE (2022). https://doi.org/10.1109/YAC57282.2022.10023749
5. Bulut, M., Özcan, E.: A new approach to determine maintenance periods of the most critical hydroelectric power plant equipment. Reliability Engineering & System Safety **205**, 107238 (2021). https://doi.org/10.1016/j.ress.2020.107238
6. Danach, K., Harb, H., Issa, H., Saker, L.: An explainable data-driven optimization framework for industrial predictive maintenance scheduling. Results in Engineering **29**, 109022 (2026). https://doi.org/10.1016/j.rineng.2026.109022
7. Darko, R., Adabor, E.S.: Computational predictive maintenance: integrating machine learning, survival analysis and optimization in dual-model framework for digital press engineering. Computers & Industrial Engineering **217**, 112043 (2026). https://doi.org/10.1016/j.cie.2026.112043
8. Delgado Sobrino, D.R., Bilacic, M., Holubek, R., Skuba, M., Felho, C., Namboodri, T.: A Framework for Integrated Maintenance of a Multi-Robot Packaging Workcell. Eng **7**(3), 134 (2026). https://doi.org/10.3390/eng7030134
9. Dui, H., Wang, H., Xing, L.: Digital Twin-Enabled Smart Operation and Maintenance Framework With Generative AI Design of Intelligent Manufacturing Systems. IEEE Transactions on Reliability **75**, 504–517 (2026). https://doi.org/10.1109/TR.2025.3646186
10. Dui, H., Wang, H., Yang, Y., Xing, L.: IoT-based mission reliability evaluation and maintenance optimization of intelligent manufacturing systems integrating human errors and heterogeneous feedstocks. Reliability Engineering & System Safety **264**, 111354 (2025). https://doi.org/10.1016/j.ress.2025.111354
11. Enjavimadar, M., Rastegar, M.: Optimal reliability-centered maintenance strategy based on the failure modes and effect analysis in power distribution systems. Electric Power Systems Research **203**, 107647 (2022). https://doi.org/10.1016/j.epsr.2021.107647
12. Gao, X., Luo, S., Peng, D., Kui, G., Xie, Y., Wu, J., Pan, J., Zuo, X., Chen, T.: A two-layer optimization method for maintenance task scheduling considering multiple priorities. Computers & Chemical Engineering **184**, 108640 (2024). https://doi.org/10.1016/j.compchemeng.2024.108640

13. Geng, S., Wang, X.: Predictive maintenance scheduling for multiple power equipment based on data-driven fault prediction. Computers & Industrial Engineering **164**, 107898 (2022). https://doi.org/10.1016/j.cie.2021.107898
14. Gopalakrishnan, M., Subramaniyan, M., Skoogh, A.: Data-driven machine criticality assessment – maintenance decision support for increased productivity. Production Planning & Control **33**(1), 1–19 (2022). https://doi.org/10.1080/09537287.2020.1817601
15. Hall, S., Gil, J., Vizcarra, O.: Boosting Runtime and Production Factors with Reliability Centered Maintenance. In: 2021 IEEE-IAS/PCA Cement Industry Conference (IAS/PCA). IEEE (2021). https://doi.org/10.1109/IAS/PCA50164.2021.10143775
16. Hedayatnia, P., Samimi, M.H., Mohammadhosein, M., Niayesh, K.: AI-enhanced reliability centered maintenance for high voltage gas circuit breakers. International Journal of Electrical Power & Energy Systems **170**, 110919 (2025). https://doi.org/10.1016/j.ijepes.2025.110919
17. Kitchenham, B., Charters, S.: Guidelines for performing systematic literature reviews in software engineering. Tech. Rep. EBSE-2007-01, Keele University and Durham University (2007)
18. Li, H., Rausell, A.L., Amiri-Simkooei, A., Ribeiro, M.: Dynamic aircraft maintenance scheduling with Graph Neural Network integrated deep reinforcement learning. Reliability Engineering & System Safety **277**, 113392 (2027). https://doi.org/10.1016/j.ress.2026.113392
19. Moradi-Sarvestani, S., Dehbozorgi, M.R., Rastegar, M.: A three-stage reliability-centered framework for critical feeder identification, failure modes prioritization, and optimal maintenance strategy assignment in power distribution system. Electric Power Systems Research **230**, 110215 (2024). https://doi.org/10.1016/j.epsr.2024.110215
20. Özcan, E., Gür, S., Eren, T.: A Hybrid Model to Optimize the Maintenance Policies in the Hydroelectric Power Plants. Journal of Polytechnic – Politeknik Dergisi **24**(1), 75–86 (2021). https://doi.org/10.2339/politeknik.626171
21. Page, M.J., McKenzie, J.E., Bossuyt, P.M., Boutron, I., Hoffmann, T.C., Mulrow, C.D., et al.: The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ **372**, n71 (2021). https://doi.org/10.1136/bmj.n71
22. Panwar, N., Kumar, S.: Critical ranking of steam handling unit using integrated cloud model and extended PROMETHEE for maintenance purpose. Complex & Intelligent Systems **7**, 367–378 (2021). https://doi.org/10.1007/s40747-020-00210-y
23. Qiu, Y., Tian, H., Chen, C., Liu, Z., Sun, Y., Li, H., Li, Y.: An improved dynamic reliability analysis method for heavy-duty CNC lathes considering maintenance. Computers & Industrial Engineering **213**, 111823 (2026). https://doi.org/10.1016/j.cie.2026.111823
24. Rios, M.P., Caiado, R.G.G., Vignon, Y.R., Corseuil, E.T., Santos, P.I.N.: Optimising Maintenance Planning and Integrity in Offshore Facilities Using Machine Learning and Design Science: A Predictive Approach. Applied Sciences **14**, 10902 (2024). https://doi.org/10.3390/app142310902
25. Rios, M.P., Kaiser, B.S., Caiado, R.G.G., Ivson, P., Roehl, D.: Decision Framework for Asset Criticality and Maintenance Planning in Complex Systems: An Offshore Corrosion Management Case. Applied Sciences **15**, 10407 (2025). https://doi.org/10.3390/app151910407
26. Schulze Spüntrup, F., Dalle Ave, G., Imsland, L., Harjunkoski, I.: Integration of maintenance scheduling and planning for large-scale asset fleets. Optimization and Engineering **23**, 1255–1287 (2022). https://doi.org/10.1007/s11081-021-09647-7
27. Shaulagara, S., Fazlurrahman, V., Riyandhini, H.P.: Risk-Based Prioritization Method for Rescheduling Delayed Preventive Maintenance of Electric Power Transmission Assets. In: 2023 4th International Conference on High Voltage Engineering and Power Systems (ICHVEPS). IEEE (2023). https://doi.org/10.1109/ICHVEPS58902.2023.10257532

28. Tang, Z., Sun, C., Ren, X., Zhang, X., Zhu, J., Li, H.: An Operating-Consistency and Evidence-Refinement Framework for Sensor-Data-Driven Photovoltaic Panel Risk Assessment and Maintenance Prioritization. Sensors **26**, 4487 (2026). https://doi.org/10.3390/s26144487
29. Wehbi, L., Bhulai, S., Slik, J.: Optimizing maintenance of heavy equipment: A data-driven approach. Computers & Industrial Engineering **211**, 111645 (2026). https://doi.org/10.1016/j.cie.2025.111645
30. Wittbrodt, P.: Algorithmic Approach to Prescriptive Maintenance in Industry 4.0. Production Engineering Archives **32**(1), 123–135 (2026). https://doi.org/10.30657/pea.2026.32.11
31. Woller, D., Rada, J., Kulich, M.: The ALNS metaheuristic for the transmission maintenance scheduling. Journal of Heuristics **29**, 349–382 (2023). https://doi.org/10.1007/s10732-023-09514-x
32. Yang, Z., Li, J., Chen, C., He, J., Tian, H., Yu, L.: A study on overall line efficiency (OLE) centered production line maintenance prioritization considering equipment operational reliability. The International Journal of Advanced Manufacturing Technology (2021). https://doi.org/10.1007/s00170-021-07547-9
33. Zhou, N., Xu, Y.: A Prioritization Method for Switchgear Maintenance Based on Equipment Failure Mode Analysis and Integrated Risk Assessment. IEEE Transactions on Power Delivery **39**(2), 728–739 (2024). https://doi.org/10.1109/TPWRD.2023.3335351